

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Department of Electrical Engineering**
2. **Subject Name: Embedded Signal Processing Laboratory, L-T-P: 0-0-3, Credits: 02**
3. **Pre-requisites: NIL**
4. **Syllabus and reference books:**

Syllabus:

Implementation of Signal Processing Algorithm (FFT, FIR, IIR Filter) on platforms like ARM 32-bit, IoT, FPGA, and GPU.

Reference Books:

- 1) Microprocessor Architecture, Programming, and Applications with the 8085 by Ramesh Gaonkar
- 2) Make: AVR Programming by Elliot Williams
- 3) STM32F103 Arm Microcontroller and Embedded Systems by Sarmad Naimi, Muhammad Ali Mazidi, and Sepehr Naimi

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Processor Familiarization with Digital IO and timers: Generate timing waveform through GPIO ports and Timer ports	Week-1
2.	Study of Aliasing: Familiarization with ADC and DAC. sample and generate a given sinusoid from the signal generator. Change the sampling frequency and see the Aliasing effect.	Week-2
3.	Interrupts and Communication: Study of interrupts and SPI communications. Generate different bit streams with interrupts and relay them through SPI or I2C. See this serial data in the DSO. The alternate bit streams could be IITKGP or DEPTTE which will be streamed alternatively with repeated occurrences with external interrupts.	Week-3
4.	Implementation of a simple low pass digital filter and studying the frequency response and transient response by feeding sinusoidal and square waves. Study the quantization error performance.	Week-4
5.	Sensor interfacing: Interface a Condenser mic to the AVR uC with proper biasing circuit and antialiasing Salen Key LPF. Speak to the mic and generate in the DSO.	Week-5
6.	Actuator interfacing: Interface a speaker with proper amplifiers. Generate the word you speak to mic with or without filtering	Week-6
	Mid-Semester Break	
7.	Familiarization with STM 32 Board: Coding in C++: direct implementation of a low pass FIR filter. Study the frequency response. Design of an IIR notch filter in MATLAB and its implementation on STM board.	Week-7
8.	Adaptive filter: Interfacing of a condenser MIC and Implementation of an LMS filter and generation of the real-time denoised signal. Input is from the condenser mic.	Week-8
9.	Familiarization with IoT: Take two modules and establish a communication by connecting them through WiFi.	Week-9
10.	Familiarization with RTOS programming on FreeRTOS and STM32 board	Week-10
11.	FPGA: Acquaintance with basic AMD-based FPGA and implementation of a simple digital circuit (FIR filter) on FPGA	Week-11
12.	GPU: Familiarization with Jetson board and writing a simple CUDA code to do FFT (DIT and DIF).	Week-12
Total number of hours		36